Field-Wise Metrics Can Miss Hierarchical Gains: Multilingual Evidence from ESG Promise Verification

Sheng-Xun Chen
National Central University
Taipei, Taiwan
tom1030507@gmail.com

Chun-Wei Chen
National Central University
Taipei, Taiwan
aaronwayway@gmail.com

Hsin-Chen Han
National Central University
Taipei, Taiwan
0720samson@gmail.com

Shi-Yu Huang
National Central University
Taipei, Taiwan
huangshiyu0318@gmail.com

ABSTRACT

ESG promise verification jointly predicts promise status, verification timeline, evidence status, and evidence quality under hard parent–child constraints. We compare seven decision rules on identical cross-fitted base probabilities and test rows in Chinese, then replicate projection versus independent argmax within English, French, Japanese, and Korean corpora. In the only confirmatory family, all five Chinese official weighted macro-F1 contrasts are undetected; projection versus argmax (M1–M0) is −0.001 with $p_{\text{Holm}} = 1.000$. Pre-specified secondary tuple accuracy instead improves by +0.035 $[0.028, 0.043]$ with $p_{\text{Holm}} = .001$. Across 32 external backbone-by-loss arms, this tuple contrast is positive in 32/32, whereas weighted macro-F1 is positive in only 14/32. Thus, in these controlled arms, field-wise scoring does not consistently reflect gains in complete hierarchical correctness.

KEYWORDS

ESG, promise verification, hierarchical classification, constrained decoding, evaluation metrics

1 INTRODUCTION

ESG promise verification is not four unrelated classification problems. The predicted promise status, verification timeline, evidence status, and evidence quality describe one claim, and their parent–child implications determine whether the combined output is internally usable. For example, a paragraph predicted to contain no promise cannot coherently receive a substantive timeline, and a prediction of no evidence cannot coherently receive an evidence quality. Individually plausible field labels can therefore form an illegal tuple.

The official weighted macro-F1 nevertheless scores each field separately. It can award credit to a substantive child whose predicted ancestors do not support that child, while a rule that preserves validity must repair or replace it. This creates a gap between what the task structure requires and what the ranking metric rewards. We ask whether the official field-wise score reflects the value of enforcing the supplied hierarchy.

We study that question as a controlled decision comparison, not as a model or competition-performance claim. Seven rules consume identical cross-fitted base probabilities and test rows. In Chinese, the confirmatory official-score family is null, yet pre-specified exact whole-tuple accuracy improves under projection. We then repeat the same within-arm M1–M0 contrast across four ML-Promise languages, four backbones per language, and two fixed training objectives. Tuple accuracy improves in all 32 external arms, while weighted macro-F1 has a mixed sign. This repeated pattern concerns decision contrasts within each corpus; raw scores do not isolate language effects.

Our contributions are:

1. We formalize the supplied hierarchy as 17 legal tuples out of 120 and audit 2,000 Chinese paragraphs from 49 companies.
2. We compare seven decision rules on shared predictions and report the complete confirmatory official-score family, the pre-specified secondary tuple family, and adverse as well as favorable results.
3. We externally replicate the within-arm projection contrast across four languages and 32 backbone-by-loss arms, while keeping post-hoc metrics and structural/backbone screens explicitly exploratory.

2 RELATED WORK

The AI CUP task uses the four-part formulation described in the PromiseEval shared-task overview: identify an ESG promise, determine whether it has actionable evidence, assess the clarity of the promise–evidence pair, and assign a verification horizon [2]. ML-Promise is the associated multilingual corpus resource; we use its English, French, Japanese, and Korean releases for external replication [9]. This joint verification problem differs from ClimateBERT-NetZero, which detects corporate and national net-zero or reduction targets and supports analysis of their ambition, rather than jointly assessing promise status, evidence, evidence quality, and timing [8].

These dependent outputs form a compact instance of hierarchical text classification, in which labels occupy a supplied structure and coherent predictions respect its ancestor relations [7]. The AI CUP hierarchy comes from the task's official field logic: we neither discover the hierarchy nor propose a new hierarchical architecture. Instead, we compare decision rules over the same field-wise probabilities while preserving or ignoring that given structure.

Yu et al. introduced path-constrained MicroF1 and MacroF1, under which a correct node receives credit only when all its ancestors are also correct [10]. Ji et al. later named this family *C-metrics* [6]. Our C-wF1 adapts Yu et al.'s principle to this field-based task and

Table 1: Dataset summary from the frozen study artifact.

Statistic	Development	Test
Paragraphs	2000	2000
Source reports (PDFs)	49	49
shared across splits	49	49
Companies	49	49
spanning >1 report	0	n/a
Legal states observed	15 / 17	n/a
Rarest classes		
within_2_years	34	n/a
Misleading	2	n/a

its official field-weighted macro-F1; it is therefore not the original C-MacroF1.

Plaud et al. observed from point estimates that constrained F1 variants did not globally change method rankings in their experiments [7]. That observation and our post-hoc paired contrast inference answer different questions: ranking a collection of models is not the same estimand as a paired difference between decision rules on shared rows and probabilities. Moreover, our hF changes both consistency treatment and macro-versus-micro aggregation. Of the two post-hoc metrics, only C-wF1 isolates the treatment of ancestor-inconsistent predictions while retaining the official field aggregation.

3 TASK AND DATA

Outputs and legal states. The task predicts four fields for each paragraph. We study the AI CUP 2026 VeriPromiseESG development release [1]. Its label inventory and parent–child logic follow PromiseEval [2]. Promise status (PS) is *Yes* or *No*. Verification timeline (VT) is *already*, *within two years*, *between two and five years*, *more than five years*, or *N/A*. Evidence status (ES) is *Yes*, *No*, or *N/A*; evidence quality (EQ) is *Clear*, *Not Clear*, *Misleading*, or *N/A*. These fields obey hard implications: PS=No requires VT=ES=EQ=N/A, whereas PS=Yes requires one of the four substantive timelines and ES in {Yes, No}. In turn, ES=No requires EQ=N/A, whereas ES=Yes requires one of the three substantive quality labels. Thus the Cartesian label space has $2 \times 5 \times 3 \times 4 = 120$ combinations, but only

$$1 + 4(1 + 3) = 17 \tag{1}$$

legal tuples: one for PS=No, and for each of four timelines either one ES=No state or three ES=Yes quality states.

Development-set audit. The release contains 2,000 labeled development paragraphs from 49 source reports and 49 companies, and it realizes 15 of the 17 legal states (Table 1). The rarest EQ label, *Misleading*, has only two instances; we therefore make no improvement or significance claim for that class.

All 49 development reports also occur in the unlabeled competition test data. Because competition test labels are unavailable, we infer no label-derived test statistic. Each company contributes one report. Consequently, in this corpus a document-disjoint split is also company-disjoint, rather than a separate company-level experiment.

External replication corpora. ML-Promise provides labels in the same four fields [9]. After normalizing its releases to the common 17-state vocabulary, we retain 400 English paragraphs from 9 reports, 400 French paragraphs from 9 reports, and 400 Japanese paragraphs from 19 reports. The Korean release supplies labels, report URLs, and page numbers but no text; we extract the corresponding 500 PDF pages from 32 reports and use the first 384 tokens of each page. The extracted Korean text is not redistributed. Korean has no gold *Misleading* example, and its page-level input is not directly comparable to the paragraph inputs. We therefore interpret only within-corpus paired decision contrasts, never raw score differences between languages.

4 METHODS

4.1 Shared Base Models

For the Chinese study, we use `hfl/chinese-roberta-wwm-ext-large` as the fixed encoder. BERT-Large has 24 layers and hidden size 1024 [4]; the pinned HFL configuration matches those dimensions. Cui et al. document the checkpoint's Chinese RoBERTa-WWM family: it uses whole-word masking, extended pretraining data, and no next-sentence prediction [3]. We mean-pool its token representations and attach four independent linear heads, one per field. The encoder is shared, but neither it nor the heads condition on predicted ancestors. The input is paragraph text only; report URL, company, ticker, and page metadata are excluded.

The encoder and matching tokenizer are loaded from the pinned snapshot revision `a25cc9e05974bd9687e528edd516f2cfdb3f5db9`.

All fits use the same frozen fine-tuning recipe: maximum length 384, 12 epochs, minibatches of 8 with two-step gradient accumulation (effective size 16), and AdamW with backbone and head learning rates 2×10^{-5} and 10^{-4}. The schedule is cosine with 10% warm-up and layer-wise decay 0.9. The four cross-entropy losses use the official PS/VT/ES/EQ weights 0.20/0.15/0.30/0.35 and label smoothing 0.1. For Train size n, K_t classes present in field t, and class support $n_{t,c}$, the class weight is $\min\{10, n/(K_t\, n_{t,c})\}$ when $n_{t,c} \geq 5$, and 1 otherwise. Hierarchy masking is disabled. We average the last three epoch checkpoints and perform no label-driven early stopping.

For the external fixed arms, English uses RoBERTa-large and French, Japanese, and Korean use XLM-R-large. English robustness runs add large DeBERTa-v3 and ELECTRA checkpoints plus RoBERTa-base. The other languages add RemBERT, XLM-R-base, and mBERT. Every backbone uses the same maximum length, optimization recipe, three seeds, and five rotations. For each backbone we run the ordinary objective ($\lambda = 0$) and a fixed hierarchy-aware objective ($\lambda = 0.3$):

$$\mathcal{L} = \mathcal{L}_{\text{fields}} - \lambda \log \sum_{s \in \mathcal{S}} \prod_t p_{t,s_t}(x). \tag{2}$$

The second term penalizes probability mass outside the 17 legal states without choosing which legal state is correct. Structural-loss and alternate-backbone comparisons are exploratory; the primary decision comparison uses $\lambda = 0$.

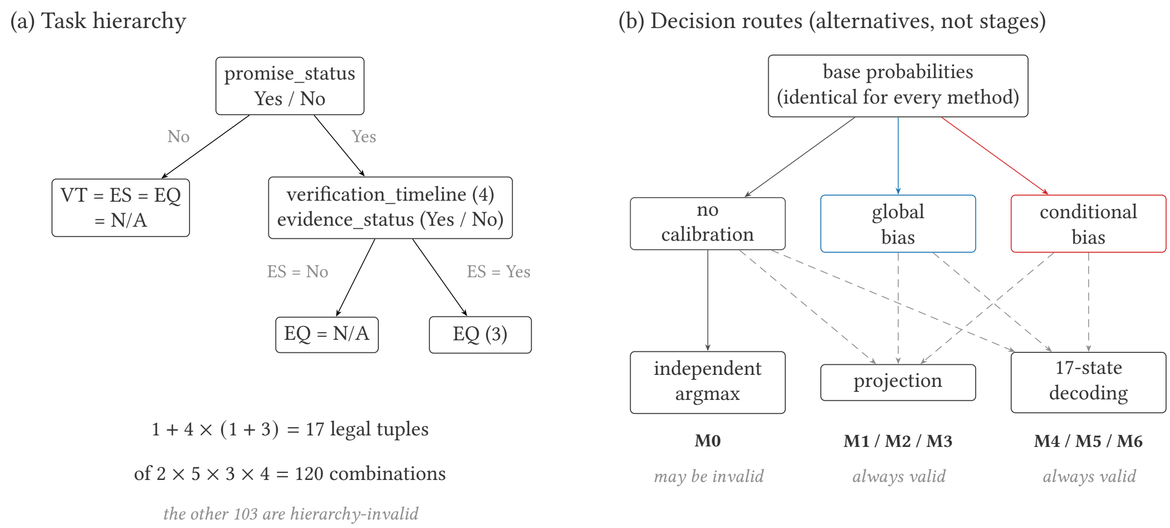

Figure 1: The task hierarchy and the alternative decision routes. All routes consume identical base probabilities; projection and 17-state decoding guarantee a legal tuple, whereas independent argmax does not.

4.2 Metric-Aware Decision Calibration

For field t and class c, decision calibration adds a class bias in log probability space:

$$z_{t,c}(x) = \log p_{t,c}(x) + b_{t,c}. \tag{3}$$

This changes decision boundaries and is not a claim of probability calibration. Each field's bias vector is fitted by deterministic coordinate ascent with that field's macro-F1 on the held-out Calibration partition as the objective, before either structured output rule is applied. The four objectives therefore decouple. Within every rotation, global and conditional biases are fitted separately; that rotation's global fit is shared by M2/M5 and its conditional fit by M3/M6.

Global biases use every Calibration row. Conditional biases use all rows for PS, only gold PS=Yes rows for VT and ES, and only gold ES=Yes rows for EQ. The three child N/A classes are structurally absent from these subsets. Their biases are therefore unidentifiable and fixed as

$$b_{\mathrm{VT,N/A}} = b_{\mathrm{ES,N/A}} = b_{\mathrm{EQ,N/A}} = 0. \tag{4}$$

This has no effect on projection, which emits N/A only when its parent requires it. It does affect joint scores for (No, N/A, N/A, N/A); hence M6−M5 compares conditional with global estimation including their different estimable parameter sets.

Biases start at zero and are fitted by deterministic coordinate ascent over $[-3, 3]$ in increments of 0.05. Coordinates follow a fixed field and class order; a move requires a strict improvement, ties favor the smaller absolute bias (then the more negative value), and search stops after a pass without improvement or at 20 passes. Eligibility is determined from the corresponding Train subset: a class absent there is excluded from coordinate ascent and retains bias zero. Among Train-eligible classes, one absent from the matching Calibration subset also retains bias zero and is recorded as an absent-class fallback.

4.3 Validity-Preserving Output Rules

Complete projection. Projection is greedy and top-down. At each field it selects the highest-scoring class allowed by the chosen ancestors. Under PS=No, it sets VT, ES, and EQ to N/A. Under PS=Yes, it excludes N/A from VT and ES and chooses each field's best substantive class. Under ES=No, EQ becomes N/A; under ES=Yes, projection chooses EQ's best substantive class. These rules completely enforce both directions of every parent implication. This bidirectional projection always returns one of the 17 legal states; a one-way N/A overwrite would not.

Joint valid-state decoding. The alternative scores every s in the legal set $\mathcal{S}$ and returns

$$\hat{y} = \arg\max_{s \in \mathcal{S}} \sum_t \alpha_t \{\log p_{t,s_t}(x) + b_{t,s_t}\}, \qquad \alpha_t = 1. \tag{5}$$

Fixing all task scales avoids giving the decoder additional tuned parameters.

Figure 1 summarizes seven *alternative* rules over identical cross-fitted base probabilities, not sequential stages: M0 is uncalibrated independent argmax; M1/M2/M3 use projection with respectively no, global, or conditional bias; and M4/M5/M6 use 17-state decoding with the same three bias choices. Only M0 can emit an illegal tuple.

5 EXPERIMENTAL DESIGN

We first define the shared evaluation protocols and inference. Table 2 then summarizes the controlled comparison interpreted in Section 6.

5.1 Cross-Fitted Evaluation Targets

We use five-way rotating three-way cross-fitting. For rotation k, Test is fold k, Calibration is fold $(k+1) \bmod 5$, and Train comprises the other three folds; the three partitions are mutually exclusive.

Table 2: Main document-disjoint results from the frozen study artifact.

ID	Calibration	Decoding	Weighted F1	PS	VT	ES	EQ	Tuple Acc.	Invalid %
M0	None	Independent	0.572±0.003	0.796	0.464	0.650	0.424	0.359	12.6
M1	None	Projection	0.571±0.003	0.796	0.467	0.651	0.418	0.394	0.0
M2	Global	Projection	0.574±0.004	0.796	0.493	0.650	0.418	0.428	0.0
M3	Conditional	Projection	0.575±0.003	0.796	0.501	0.651	0.416	0.432	0.0
M4	None	17-state	0.571±0.005	0.799	0.468	0.648	0.420	0.388	0.0
M5	Global	17-state	0.576±0.001	0.796	0.493	0.649	0.423	0.430	0.0
M6	Conditional	17-state	0.567±0.008	0.784	0.494	0.634	0.418	0.427	0.0

After five rotations, every development paragraph has served as Test exactly once. We concatenate the 2,000 out-of-fold predictions and compute each metric once, rather than average fold-level F1 values whose present gold-label sets may differ. The three seeds 42, 123, and 456 each determine both a fold assignment and training randomness, so seed variation represents the complete pipeline.

The primary, document-disjoint protocol balances report row counts with a randomized greedy procedure. After a seeded shuffle, it processes reports from largest to smallest and randomly places each in one of the two currently lightest folds. It estimates unseen-report generalization and, given the corpus audit, is also company-disjoint.

The same-document protocol instead jointly stratifies by gold PS, ES, and VT. It shuffles rows within each joint-label bucket and deals them round-robin to the five folds. Every report represented in Calibration or Test must also occur in Train through a different paragraph. This estimates seen-report, unseen-paragraph generalization. The protocols are distinct estimands, not biased and unbiased versions of one quantity.

Within each protocol and seed, M0–M6 consume the same Test rows and base probabilities.

External runs use document-disjoint report grouping, three seeds, and five rotations. Each language has eight paired arms: four backbones under both $\lambda = 0$ and 0.3, giving 32 external arms. The fixed-arm table uses $\lambda = 0$: RoBERTa-large for English and XLM-R-large for the other three languages. Arm choice never depends on observed scores. The same-document protocol is descriptive and available only for the primary large backbone.

5.2 Metrics and Inference

For Chinese, the primary metric is the official weighted macro-F1,

$$\mathrm{wF1} = 0.20F_{\mathrm{PS}} + 0.15F_{\mathrm{VT}} + 0.30F_{\mathrm{ES}} + 0.35F_{\mathrm{EQ}}, \tag{6}$$

where each F_t is field-level macro-F1 over classes present in the gold labels being scored. Because this denominator depends on gold support, per-class leverage differs by corpus: a unit gain on one EQ class contributes $0.35/3 = 0.1167$ to the Korean weighted total but $0.35/4 = 0.0875$ elsewhere. Pre-specified whole-tuple accuracy requires all four fields to match. Path-constrained weighted macro-F1 (C-wF1) and hierarchical F1 (hF) were adopted post hoc. C-wF1 retains the official field weights but counts a substantive child prediction as wrong when its predicted ancestors do not support it. Following the hierarchical precision/recall/F1 set-overlap definition [7], our hF micro-averages over emitted non-N/A field–value nodes: hP is total gold–prediction overlap divided by predicted nodes, hR divides the same overlap by gold nodes, and hF is their harmonic mean. For an inconsistent emitted tuple, we do not add missing ancestors before scoring; legal M1–M6 are unaffected because the conventions coincide for them. hF changes both consistency treatment and macro-versus-micro averaging, so the two supplementary metrics answer different questions.

ML-Promise has no official weighted metric. We therefore label its scores "AI CUP weights applied to ML-Promise." All external comparisons remain paired within corpus, backbone, and λ.

For paired method contrasts, we perform a paired PDF-cluster bootstrap over 49 source-report clusters. Each of 10,000 replicates samples 49 reports with replacement using seed 20260814, applies the same sampled rows to both methods and all three seeds, computes the within-seed differences, and averages those differences. The five pre-specified contrasts are M1–M0, M3–M2, M4–M1, M6–M3, and M6–M5. Our only confirmatory family applies Holm correction to these five contrasts on Chinese official wF1 [5]. Pre-specified Chinese tuple exact-match is a separate secondary family of five. Bracketed 95% intervals are uncorrected bootstrap percentile intervals; decisions use $p_{\mathrm{Holm}} < .05$, not whether an interval excludes zero.

External replication was pre-registered around within-arm contrast directions; we additionally apply a report-cluster bootstrap only to the fixed English M1–M0 tuple contrast. We do not pool languages into a post-hoc significance test. C-wF1 and hF were added after the Chinese primary result, while structural-loss and alternate-backbone screens test robustness; all are exploratory and do not support confirmatory claims regardless of their nominal p-values.

6 RESULTS

Chinese confirmatory family. Table 2 compares decision rules on identical base probabilities and Test rows. Independent argmax (M0) produces invalid tuples for 12.55% of document-disjoint predictions; projection and 17-state decoding (M1–M6) reduce this rate to zero by construction. The ± values are sample standard deviations across the three seeds. Because each seed determines both fold assignment and training randomness, they describe spread over the whole split-and-training pipeline, not a confidence interval.

Table 3: Chinese paired contrasts on four metrics. Official wF1 and tuple accuracy are separate pre-specified Holm families of five. C-wF1 and hF were added post hoc; their displayed adjusted values are exploratory and support no claim. Intervals are uncorrected 95% report-cluster bootstrap intervals.

Metric	Contrast	Δ	95% CI	p_{Holm}
Weighted macro-F1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.416
	M6-M5	-0.008	[-0.016, -0.001]	0.171
Path-constrained wF1	M1-M0	+0.004	[0.001, 0.007]	**0.025**
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.312
	M6-M5	-0.008	[-0.016, -0.001]	0.137
Hierarchical F1 (hF)	M1-M0	+0.003	[0.001, 0.005]	**0.017**
	M3-M2	+0.002	[-0.001, 0.006]	0.286
	M4-M1	+0.004	[0.000, 0.007]	0.135
	M6-M3	+0.006	[-0.001, 0.012]	0.224
	M6-M5	+0.004	[-0.001, 0.009]	0.286
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.004	[-0.001, 0.008]	0.370
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.015, 0.004]	0.505
	M6-M5	-0.004	[-0.011, 0.004]	0.505

Table 4: Fixed document-disjoint, lambda=0 M1–M0 contrasts. Rates and deltas are means over 3 seeds; the net columns report repairs minus destroyed field predictions pooled over those seeds. Chinese uses Chinese RoBERTa WWM Ext Large, English uses RoBERTa-large, and French, Japanese, and Korean use XLM-R-large. M0 invalid is the independent-argmax invalid-tuple rate. For ML-Promise, weighted F1 means AI CUP weights applied to ML-Promise. Korean uses PDF-page inputs and has zero gold Misleading examples (n=0), so one evidence-quality class-F1 point contributes 0.1167 there versus 0.0875 elsewhere.

Language (corpus)	Rows	M0 invalid %	*N/A* net	Subst. net	Δ wF1	Δ tuple
Chinese (AI CUP)	2,000	12.55%	+395	-264	-0.0011	+0.0350
English (ML-Promise)	400	23.17%	+148	-86	+0.0032	+0.0725
French (ML-Promise)	400	21.08%	+150	-107	+0.0036	+0.0625
Japanese (ML-Promise)	400	31.25%	+183	-245	-0.0131	+0.0708
Korean (ML-Promise)	500	22.73%	+103	-148	-0.0192	+0.0293

All five official wF1 contrasts are undetected after Holm correction. For the central M1–M0 contrast, $\Delta = -0.001$ with 95% interval $[-0.006, 0.003]$ and $p_{\mathrm{Holm}} = 1.000$. This is absence of a detected difference; it does not establish a zero effect.

Pre-specified tuple family. Exact whole-tuple accuracy tells a different story. M1–M0 is +0.035 $[0.028, 0.043]$, $p_{\mathrm{Holm}} = .001$. Conversely, joint legal-state decoding is not uniformly better: M4–M1 is -0.006 $[-0.010, -0.002]$, $p_{\mathrm{Holm}} = .028$. Searching every legal state can therefore reduce whole-row correctness relative to greedy projection.

The post-hoc C-wF1 and hF M1–M0 point estimates are also positive (+0.004 and +0.003), but they do not alter the inferential hierarchy. In the same-document sensitivity analysis, C-wF1 remains positive at +0.002, while its interval $[-0.001, 0.005]$ crosses zero.

Multilingual replication. Table 4 fixes one arm per language before inspecting scores. Each uses document-disjoint evaluation and $\lambda = 0$.

In these fixed arms, projection has a positive *N/A* net and a negative substantive-class net in all five corpora. Whole-tuple accuracy rises in all five, whereas weighted macro-F1 rises only for English and French and falls for the other three.

Across all four external corpora and their four backbones at both loss settings, M1–M0 tuple accuracy is positive in 32/32 arms. The weighted macro-F1 contrast is positive in only 14/32: English 7/8, French 5/8, Japanese 0/8, and Korean 2/8. These are within-arm directions, not a raw score ranking across languages. For the pre-registered fixed English arm, the one-sided report-cluster bootstrap tuple effect is +0.0725 $[0.0452, 0.1019]$ ($p = .0002$; nine report clusters). ML-Promise weighted scores use AI CUP weights applied to ML-Promise.

Exploratory robustness. On the Chinese anchor, training-time structural loss lowers M0 invalidity from 12.55% to 5.18%, but its pre-registered M1 wF1 effect is only +0.00250 $[-0.00384, 0.00860]$ ($p = .427$). Across the 16 external backbone pairs, structural loss lowers invalidity in 16/16, while downstream wF1 and tuple effects are mixed. Chinese DeBERTa and ELECTRA screens likewise reduce invalidity without a conclusive wF1 gain. Chinese RoBERTa-base emits fewer, not more, invalid tuples than the large anchor, falsifying the simple smaller-model mechanism. These screens are descriptive and exploratory.

7 DISCUSSION

The repeated result is a contrast pattern, not a ranking of languages or raw scores. Projection enforces the supplied hierarchy in every arm and improves exact whole-tuple accuracy throughout the executed external matrix. Its effect on field-wise weighted macro-F1 instead changes sign across backbones and corpora. Together with the Chinese confirmatory null and pre-specified tuple gain, this supports a metric–structure mismatch in these controlled arms: a field-weighted score need not reward a decision rule for making the combined output more coherent and more often exactly correct.

The measured ledger makes the trade-off explicit: the *N/A* net is positive and the substantive-class net negative in all five fixed arms. If an independently predicted parent is wrong, projection may replace a correct substantive child with the structurally required *N/A*; when the parent is correct, the same operation can repair an unsupported child. Field-wise macro-F1 values these changes class by class, whereas tuple accuracy credits only four-field agreement. Post-hoc C-wF1 and hF point in the tuple direction, but were chosen after the primary analysis and are not needed to establish the pre-specified secondary result.

Validity alone also does not identify the better structured rule. In Chinese, M4 scores all 17 legal states jointly yet loses tuple accuracy to greedy M1. Joint search can improve one field while changing another field that projection had already made correct. This adverse result cautions against treating a larger legal-state search as automatically superior; it does not imply that joint decoding must fail on other models or tasks.

7.1 Limitations

The corpora differ in language, reports, annotation distributions, and primary backbones, so the replication matrix is not a factorial estimate of a pure language effect. English and French each have only nine report clusters, and even Chinese inference resamples just 49. The 32/32 and 14/32 summaries are therefore descriptive directions across executed arms, not a pooled significance test. AI CUP weights applied to ML-Promise are not an official ML-Promise metric.

Korean has a further input mismatch: its 500 examples are first-384-token PDF pages rather than released paragraphs, and it has no gold *Misleading* example. The extracted local page text and source PDFs are intentionally not redistributed. Nevertheless, the committed split manifests and row-level predictions are sufficient to rescore the reported Korean decision results from a fresh clone. They are not sufficient to retrain the models without reconstructing the local page inputs.

Results for structural loss and other backbones remain exploratory. They show that reducing invalidity need not yield a conclusive gain on the official score, but the fixed loss weights and limited checkpoint set cannot characterize all training objectives or architectures. The Chinese rarest quality label has only two examples, Korean has none, and no class-level claim is licensed for it. Finally, each seed jointly changes folds and optimization randomness, so those sources of variation cannot be separated.

8 CONCLUSION

Across the controlled Chinese study and four external corpora, field-wise weighted macro-F1 did not consistently reflect the validity and exact-match gains observed after hierarchical projection. Evaluations of dependent output fields should therefore report both field-wise metrics and a structured whole-output measure, while treating validity as a requirement rather than a surrogate for downstream accuracy.

REFERENCES

[1] AI CUP 2026 Organizers. 2026. ESG Sustainability Commitment Verification Competition 2026 (VeriPromiseESG4K). AIdea competition website. https://www.aidea-web.tw/aicup_veripromiseesg

[2] Chung-Chi Chen, Yohei Seki, Hakusen Shu, Anaïs Lhuissier, Juyeon Kang, Hanwool Lee, Min-Yuh Day, and Hiroya Takamura. 2025. SemEval-2025 Task 6: Multinational, Multilingual, Multi-Industry Promise Verification. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Association for Computational Linguistics, Vienna, Austria, 2461–2471. https://aclanthology.org/2025.semeval-1.321/

[3] Yiming Cui, Wanxiang Che, Ting Liu, Bing Qin, and Ziqing Yang. 2021. Pre-Training With Whole Word Masking for Chinese BERT. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021), 3504–3514. https://doi.org/10.1109/TASLP.2021.3124365

[4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. https://doi.org/10.18653/v1/N19-1423

[5] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. https://www.jstor.org/stable/4615733

[6] Ke Ji, Yixin Lian, Jingsheng Gao, and Baoyuan Wang. 2023. Hierarchical Verbalizer for Few-Shot Hierarchical Text Classification. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Toronto, Canada, 2918–2933. https://doi.org/10.18653/v1/2023.acl-long.164

[7] Roman Plaud, Matthieu Labeau, Antoine Saillenfest, and Thomas Bonald. 2024. Revisiting Hierarchical Text Classification: Inference and Metrics. In *Proceedings of the 28th Conference on Computational Natural Language Learning*. Association for Computational Linguistics, Miami, FL, USA, 231–242. https://doi.org/10.18653/v1/2024.conll-1.18

[8] Tobias Schimanski, Julia Bingler, Mathias Kraus, Camilla Hyslop, and Markus Leippold. 2023. ClimateBERT-NetZero: Detecting and Assessing Net Zero and Reduction Targets. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 15745–15756. https://doi.org/10.18653/v1/2023.emnlp-main.975

[9] Yohei Seki, Hakusen Shu, Anaïs Lhuissier, Hanwool Lee, Juyeon Kang, Min-Yuh Day, and Chung-Chi Chen. 2025. ML-Promise: A Multilingual Dataset for Corporate Promise Verification. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. https://aclanthology.org/2025.emnlp-main.1028/

[10] Chao Yu, Yi Shen, and Yue Mao. 2022. Constrained Sequence-to-Tree Generation for Hierarchical Text Classification. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Madrid, Spain). Association for Computing Machinery, New York, NY, USA, 1865–1869. https://doi.org/10.1145/3477495.3531765